



Multilayer high precision PCB's with impedance control

JLCPCB can produce High-precision multilayer board with capabilities listed in below table.

Min. Trace width/Spacing	Min. Via	Min. BGA
3.5mil	0.2mm	0.25mm

It is very convenient for customers to conduct an impedance matching design, according to JLCPCB's laminated structure and related parameters. The PCB will be accordance with the following stackup.

Controlled Impedance PCB Parameters and Stackup

1.Prepreg dielectric constant:

Prepreg type	Dielectric constant
7628	4.4
3313	4.1
1080	3.91
2116	4.16

2.Solder mask Parameters

Coating Above Substrate C1	Coating Above Trace C2	Coating Between Traces C3	Coating Dielectric CEr
1.2mil	0.6mil	1.2mil	3.8

3. Core dielectric constant

Core dielectric constant
4.6

For your convenience, we have designed an [Impedance Calculator](#) to help you calculate the impedance and the trace width you require.

4-Layer Impedance Control Stackup

Thickness

0.8mm	1.0mm	1.2mm	1.6mm	2.0mm
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Outer Copper Weight

1oz	2oz
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inner Copper Weight

0.5oz	1oz	2oz
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1) No requirement Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	7628*1	0.203mm	
Inner Layer L2	Copper	0.03mm	1.1mm 1/10Z with copper
Core	Core	1.03mm	
Inner Layer L3	Copper	0.03mm	
Prepreg	7628*1	0.203mm	
Bottom Layer	Copper	0.035mm	

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2) JLC041611-7628 Stackup

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